

TOM (TIANZE) YANG

tianze.yang@mail.mcgill.ca | (+1) 514-560-9523 | [LinkedIn](#) | [GitHub](#)

SUMMARY

- Experience: AI software engineer with production agentic systems, multimodal/voice agents, ML research, and full-stack engineering experience
- Languages: Python, TypeScript, SQL, Bash, R, JavaScript; fluent in English and Chinese
- Technical Skills: Agentic systems, tool/function calling, tool registry/MCP, memory/context management, structured outputs, AI-native engineering, STT/TTS, RAG, QLoRA/DPO, PyTorch/TensorFlow, FastAPI, PostgreSQL, Docker, AWS EventBridge/Lambda, Terraform, React/Next.js

EDUCATION

McGill University

M.Sc.(Thesis) in Computer Science, Supervisor: [Prof. Xue Liu](#)

- GPA: 3.92/4.0

B.Sc. in Honors Probability and Statistics (DS & ML Path), Minor Computer Science

- GPA: 3.85/4.0

Montreal, Canada

Sep 2023 - Sep 2025

Sep 2020 - May 2023

WORK EXPERIENCE

Palona AI

Montreal, Canada

Founding Engineer, Research & AI Software

Mar 2024 - present

- Joined at inception to turn early research prototypes into customer-facing AI software, helping build the product and engineering foundation through the company's \$10M seed round
- Built a voice ordering agent across STT-to-LLM-to-TTS and speech-to-speech architectures, integrating function calling, multilingual TTS, and per-client knowledge bases for production restaurant workflows
- Improved skill routing with recommendation injection, lifting correct skill invocation by +29-35pp with zero false positives and 95% router accuracy.
- Created a voice-agent evaluation and self-improvement loop that converted real conversation failures into targeted prompt, context, and workflow updates
- Implemented structured decoding and function calling for LLM workflows, making tool/API use more reliable in multi-step customer interactions
- Built event-driven multi-tenant SMS infrastructure with EventBridge, Lambda, and Terraform, supporting outbound number mapping, routing, and client isolation
- Established AI-assisted engineering workflow standards for context gathering, code review, and failure analysis, reducing ambiguity as the engineering team scaled

Nebula AI

Montreal, Canada

AI Engineer Intern

Sep 2023 - May 2024

- Won runner-up in the IPC Data Economy Hackathon by developing and deploying an IPFS NSFW content filter with Vision Transformer, PyTorch, Flask, and Docker
- Built stroke-prediction models using ensemble methods (Transformers, Random Forests, Decision Trees), Fisher-score feature selection, and QLoRA fine-tuning, reaching top 5% performance on the benchmark
- Created a generative art NFT system that combined Chainlink Functions, Stable Diffusion, and multi-chain storage to support on-chain AI media workflows

McGill University & MILA

Montreal, Canada

Research Intern. Mentors: Dr. Shaoshan Liu, Prof. Xue Liu

Sep 2022 - Sep 2025

- Developed ICE-SEARCH to automate feature-selection workflows with LLM prompting, Transformers, and Python, producing a language-model-driven data engineering approach
- Built AICOM, a TensorFlow monkeypox screening system that achieved >98% accuracy and >1000 downloads, then deployed it to resource-constrained Android devices with float-16 quantization
- Open-sourced a web-based screening system with Angular, Flask, and Google Cloud, entering the project into the MIT Solve challenge for health-AI access
- Computed Dimson Beta across 873K+ stock observations and modeled Amihud illiquidity/systematic volatility risk using regression-based quantitative workflows

Shenzhen Institute of Computing Science

Shenzhen, China

Research Intern. Mentor: Dr. Weijie Ou

Jul 2023 - Sep 2023

- Fine-tuned Llama2-7B for NL2SQL with QLoRA, improving execution accuracy by ~3x on YashanDB query-generation tasks
- Improved NL2SQL accuracy by ~2.2x through Chain-of-Thought, few/zero-shot, and role-play prompting experiments
- Implemented a Python/Bash NL2SQL engine for YashanDB, validating natural-language parsing against executable SQL behavior

SIDE PROJECTS

Gstrl - macOS gesture and voice control app using Apple Vision hand tracking + on-device speech recognition (Swift)

Loupe - Real-time observer for Claude Code sessions, streaming tool calls and agent activity into TUI + native macOS window (JavaScript)

Dagashi - macOS app that gamifies typing with anime ASCII art gacha (Swift/Tauri)

PUBLICATIONS

- [Development and Deployment of Jami: Experiences with a Mobile Distributed Communication Platform](#)
Yili Jin, Tianyi Yang, Adrien Béraud, **Tianze Yang**, Sébastien Blin, Cyrille Béraud, Qiao Xiang, Xue Liu
IEEE TMC
- [ICE-SEARCH: A Language Model-Driven Feature Selection Approach \(co-First Author\)](#)
Tianze Yang*, Tianyi Yang*, Shaoshan Liu, Fuyuan Lyu, and Xue Liu
ArXiv
- [Achieving United Nations' SDG3 Through Empowering Health Artificial Intelligence on Resource-Constrained Mobile Devices Without Connectivity \(co-First Author\)](#)
Tim Tianyi Yang*, **Tom Tianze Yang***, Shaoshan Liu, and Xue Liu
ACM SIGCAS
- [Keep Your Paths Free: Toward Scalable Learning-Based Traffic Engineering](#)
Haoyu Liu, Siyong Huang, Shaoxiang Qin, **Tianze Yang**, Tianyi Yang, Qiao Xiang, Xue Liu
APNet '24
- [AICOM-MP: an AI-based Monkeypox Detector for Resource-Constrained Environment \(co-First Author\)](#)
Tim Tianyi Yang*, **Tom Tianze Yang***, Andrew Liu, Jie Tang, Na An, Shaoshan Liu, and Xue Liu
Connection Science
- [AI Clinics on Mobile \(AICOM\): Universal AI Doctors for the Underserved and Hard-to-Reach \(co-First Author\)](#)
Tim Tianyi Yang*, **Tom Tianze Yang***, Na An, Ao Kong, Shaoshan Liu, and Xue Liu
ArXiv

HONORS

- Runner-up in the Filecoin Track of the InterPlanetary Consensus (IPC): Data Economy Hackathon (2024)
- First-Class-Honors in Probability and Statistics (2023)
- William Macdonald Scholarship (2020-2023)
- Research Assistant Stipend at XiaLab (2021-2023)
- Research Assistant Stipend at Weichenthal's Lab (2021)
- McGill Science Dean's Honor List (2020-2021)
- Wing Hing Chan Scholarships in Science (2020-2021)
- McGill Emerging Leaders Certificate (2021)
- Governor General of Canada, Bronze (2018)